STACKS, MODULI OF CURVES: LEMMA-UNIQUE-STABLE-MODEL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3272-3281 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 3283-3376, the complete author proof (92 body lines).

Editorial note. Navigation stubs: 16 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 2 of queue_moduli-curves.json (pb=4146, stmt 3272-3281, proof 3283-3376).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E7Q: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E89: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 2900–2912. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E8A: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3127–3138. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CDC: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CDF: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CDE: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C6B: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CD9: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CDG: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C2W: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C2R: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7F: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7G: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C2I: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3376–3376. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let R be a discrete valuation ring with fraction field K. Let C be a smooth proper curve over K with $K = H^0(C, \mathcal{O}_C)$ and genus g. If X and X' are models of C (Semistable Reduction, Section 0C2R) and X and X' are stable families of genus g curves over R, then there exists a unique isomorphism $X \to X'$ of models.*

Proof. Let Y be the minimal model for C. Recall that Y exists, is unique, and is at-worst-nodal of relative dimension 1 over R, see Semistable Reduction, Proposition 0C2W and Lemmas 0C6B and 0CDG (applies because we have X). There is a contraction morphism

$$Y \longrightarrow Z$$

such that Z is a stable family of curves of genus g over R (Lemma 0E8A). We claim there is a unique isomorphism of models $X \to Z$. By symmetry the same is true for X' and this will finish the proof.

By Semistable Reduction, Lemma 0CDE there exists a sequence

$$X_m \to \ldots \to X_1 \to X_0 = X$$

such that $X_{i+1} \to X_i$ is the blowing up of a closed point x_i where X_i is singular, $X_i \to \mathrm{Spec}(R)$ is at-worst-nodal of relative dimension 1, and X_m is regular. By Semistable Reduction, Lemma 0CD9 there is a sequence

$$X_m = Y_n \to Y_{n-1} \to \ldots \to Y_1 \to Y_0 = Y$$

of proper regular models of C, such that each morphism is a contraction of an exceptional curve of the first kind[1]. By Semistable Reduction, Lemma 0CDF each Y_i is at-worst-nodal of relative dimension 1 over R. To prove the claim it suffices to show that there is an isomorphism $X \to Z$ compatible with the morphisms $X_m \to X$ and $X_m = Y_n \to Y \to Z$. Let $s \in \mathrm{Spec}(R)$ be the closed point. By either Lemma 0E89 or Lemma 0E8A we reduce to proving that the morphisms $X_{m,s} \to X_s$ and $X_{m,s} \to Z_s$ are both equal to the canonical morphism of Algebraic Curves, Lemma 0E7Q.

For a morphism $c : U \to V$ of schemes over $\kappa(s)$ we say c has property (*) if $\dim(U_v) \leq 1$ for $v \in V$, $\mathcal{O}_V = c_*\mathcal{O}_U$, and $R^1c_*\mathcal{O}_U = 0$. This property is stable under composition. Since both X_s and Z_s are stable genus g curves over $\kappa(s)$, it suffices to show that each of the morphisms $Y_s \to Z_s$, $X_{i+1,s} \to X_{i,s}$, and $Y_{i+1,s} \to Y_{i,s}$, satisfy property (*), see Algebraic Curves, Lemma 0E7Q.

Property (*) holds for $Y_s \to Z_s$ by construction.

The morphisms $c : X_{i+1,s} \to X_{i,s}$ are constructed and studied in the proof of Semistable Reduction, Lemma 0CDE. It suffices to check (*) étale locally on $X_{i,s}$. Hence it suffices to check (*) for the base change of the morphism "$X_1 \to X_0$" in Semistable Reduction, Example 0CDC to $R/\pi R$. We leave the explicit calculation to the reader.

The morphism $c : Y_{i+1,s} \to Y_{i,s}$ is the restriction of the blow down of an exceptional curve $E \subset Y_{i+1}$ of the first kind, i.e., $b : Y_{i+1} \to Y_i$ is a contraction of E, i.e., b is a blowing up of a regular point on the surface Y_i (Resolution of Surfaces, Section 0C2I). Then $\mathcal{O}_{Y_i} = b_*\mathcal{O}_{Y_{i+1}}$ and $R^1b_*\mathcal{O}_{Y_{i+1}} = 0$, see for example Resolution of Surfaces, Lemma 0AGS. We conclude that $\mathcal{O}_{Y_{i,s}} = c_*\mathcal{O}_{Y_{i+1,s}}$ and $R^1c_*\mathcal{O}_{Y_{i+1,s}} = 0$ by More on Morphisms, Lemmas 0E7F, 0E7G, and 0E7I (only gives surjectivity of $\mathcal{O}_{Y_{i,s}} \to c_*\mathcal{O}_{Y_{i+1,s}}$ but injectivity follows easily from the fact that $Y_{i,s}$ is reduced and c changes things only over one closed point). This finishes the proof. □

[1] In fact we have $X_m = Y$, i.e., X_m does not contain any exceptional curves of the first kind. We encourage the reader to think this through as it simplifies the proof somewhat.